Curriculum for Undergraduate Degree (B.Tech.) in Automobile Engineering (w.e.f. AY: 2020-21)

Part III: Detailed Curriculum

Third Semester

Course Name:	**Mathematics -III**		
Course Code:	BS-M303	**Category:**	BS
Semester:	Third	**Credit:**	3
L-T-P:	2-1-0	**Pre-Requisites:**	High school mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To understand probability theory and its applications.
2	To know the concept of Complex Analysis.
3	To introduce the solution methodologies for second order Partial Differential Equations with applications in engineering.
4	Learn different tools of differentiation and integration of functions of a complex variable that are used with various other techniques for solving engineering problems.
5	To provide an overview of statistics to engineers.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Basic Probability:** • **Probability** (i) Definition of random experiment, sample space, events and probability. (ii) Basic theorems (Statement only) of probability. (iii) Conditional probability and independent events; Multiplication theorem; Baye's theorem (statement only) and related problems. • **Probability Distribution**	**10**

	(i) Definition of random variable; Discrete and continuous random variable; Probability mass function (p.m.f.) and probability density function (p.d.f.) of single random variable; Cumulative distribution function (c.d.f.); Applications. (ii) Expectation and variance of random variable; Properties and applications. **(iii) Some special types of distributions** ➢ **Discrete probability distribution:** Binomial and Poisson distributions; Mean and variance (no proof) and examples. ➢ **Continuous probability distribution:** Uniform, Exponential and Normal distributions; Mean and variance (no proof) and examples.	
2	**Statistics:** • **Measure of Central Tendency** (i) Statistical data and frequency distribution. (ii) Mean, Median and Mode (formulae only) and related problems. (iii) Variance and standard deviation (formulae only) and applications. • **Regression Analysis** (i) Introduction to bivariate data; Scatter diagram. (ii) Correlation and Correlation Coefficient, Rank Correlation; related problems. (iii) Regression line and linear curve fitting; Properties of regression line and coefficients; related problems. (iv) Introduction to non-linear regression.	**6**
3	**Calculus of Complex Variables:** • **Introduction to differential calculus of function of complex**	**12**

	variable (i) Function of complex variable. (ii) Concept of Limit, continuity and differentiability. (iii) Analytic function; Cauchy-Riemann equations (Statement only); Sufficient conditions for a function to be analytic; Harmonic function and Conjugate Harmonic function; related problems. (iv) Construction of Analytic function; Milne-Thomson Method; related problems. • **Complex Integral Calculus** (i) Zeros and singularities of an analytic function: Zeros of an analytic function; Singularities of an analytic function, Nature and Location of Singularities, Pole; Examples. (ii) Concept of simple curve, closed curve, smooth curve and contour; Line integrals along a piecewise smooth curve; Examples. (iii) Cauchy's Theorem (statement only), Cauchy-Gousrsat Theorem (statement only), Examples. (iv) Cauchy's Integral Formula; examples. (v) Taylor's series, Laurent's series; examples. (vi) Residues of a given function. (vii) Cauchy's Residue Theorem (statement only); evaluation of definite integrals involving sine and cosine.	
4	**Bessel and Legendre Equations:** • **Bessel's Equation** (i) Series solution of Bessel's equation. (ii) Bessel's function; Recurrence relations of Bessel's function of first kind; Examples.	**6**

	• **Legendre's Equation** (i) Series solution of Legendre's equation. (ii) Legendre's Polynomials. (iii) Generating function of Legendre Polynomials and Orthogonal Properties; Examples. (iv) Recurrence relations; Examples. (v) Rodrigue's Formula; Examples.	
5	**Solution of Partial Differential Equations:** (i) Brief introduction to PDEs; Types of PDEs. (ii) Solution of Boundary Value Problems by Method of Separation of Variables: (a) Two dimensional Laplace equation (b) One dimensional heat conduction equation (c) One dimensional wave equation	**6**
Total		**40**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Learn the ideas of probability and random variables, various discrete and continuous probability distributions with their properties and their applications in physical and engineering environment.
2	To apply statistical methods for analysing experimental data.
3	Apply statistical tools for analysing complex field.
4	Students will be able to solve field problems in engineering involving PDEs.

Learning Resources:	
1	Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons.
2	Michael Greenberg, Advanced Engineering Mathematics, Pearson.
3	B.S. Grewal, Higher Engineering Mathematics, Khanna Publishers.
4	Kanti B. Dutta, Mathematical Methods of Science and Engineering, Cenage Learning
5	Reena Garg, Chandrika Prasad, Advanced Engineering Mathematics, Khanna Publishers.
6	N.G. Das, Statistical Methods (Combined Volume), Tata-McGraw Hill
7	S. Ross, A First Course in Probability, Pearson Education India
8	W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, Wiley
9	N.P. Bali and Manish Goyal, A text book of Engineering Mathematics, Laxmi Publications, Reprint, 2010
10	Advanced differential equation-M.D. Raisinghania, S. Chand Publication

Course Name:	**Biology**		
Course Code:	BS-BIO301	**Category:**	Basic Science Course
Semester:	Third	**Credit:**	2
L-T-P:	2-0-0	**Pre-Requisites:**	Basic knowledge of Physics, Chemistry and Mathematics
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	Bring out the fundamental differences between science and engineering
2	Discuss how biological observations of 18th Century that lead to major discoveries

Course Contents:		
Module No.	**Description of Topic**	**Contact Hrs.**

1	**Module 1- Introduction to Biology:** To convey that Biology is as important a scientific discipline as Mathematics, Physics and Chemistry Bring out the fundamental differences between science and engineering by drawing a comparison between eye and camera, Bird flying and aircraft. Mention the most exciting aspect of biology as an independent scientific discipline. Why we need to study biology? Discuss how biological observations of 18th Century that lead to major discoveries. Examples from Brownian motion and the origin of thermodynamics by referring to the original observation of Robert Brown and Julius Mayor. These examples will highlight the fundamental importance of observations in any scientific inquiry.	2
2	**Module2-Classification System in Biology:** The underlying criterion, such as morphological, biochemical or ecological be highlighted. Hierarchy of life forms at phenomenological level. A given organism can come under different category based on classification. Model organisms for the study of biology come from different groups. *E. coli, S. cerevisiae, D. melanogaster, C. elegance, A. thaliana, M. musculus.*	2
3	**Module 3: Genetics:** To convey that "Genetics is to biology what Newton's laws are to Physical Sciences" Mendel's laws, Concept of segregation and independent assortment. Concept of allele. Gene mapping, Geneinteraction, Epistasis. Meiosis and Mitosis be taught as a part of genetics.Emphasis to be given not to the mechanics of cell division nor the phases but how genetic material passes from parent to offspring. Importance of stem cell research.	2
4	**Module 4: Biomolecules**: To convey that all forms of life have the same building blocks and yet the manifestations are as diverse as one can imagine Molecules of life. Inthis context discuss monomeric units and polymeric structures. Discuss about sugars, starch and cellulose. Amino acids and proteins. Nucleotides and DNA/RNA.	4
5	**Module 5: Enzymes:** To convey that without catalysis life would not have existed on earth Enzymology: How to monitor enzyme catalysed reactions. How does an enzyme catalyse reactions? Discuss at least two examples.	2
6	**Module 6: Information Transfer:** The molecular basis of coding and decoding genetic information is universal Molecular basis of information transfer. DNA as a genetic material. Hierarchy of DNA structure- from single stranded to double helix to nucleosomes. Concept of genetic code. Universality and degeneracy of genetic code. Define gene in terms of complementation and recombination.	4

7	**Module 7: Macromolecular analysis:** How to analyse biological processes at the reductionist level Proteins- structure and function. Hierarch in protein structure. Primary secondary, tertiary and quaternary structure. Proteins as enzymes, transporters, receptors and structural elements.	4
8	**Module 8: Metabolism:** ATP as an energy currency. This should include the breakdown of glucose to CO2 + H2O (Glycolysis and Krebs cycle) and synthesis of glucose from CO2 and H2O (Photosynthesis). Energy yielding and energy consuming reactions. Concept of Energy charge.	2
9	**Module 9: Microbiology:** Concept of microscopic organisms. Concept of species and strains. Identification and classification of microorganisms. Sterilization and media compositions. Growth kinetics. Microscopy: simple, compound, phase-contrast, SEM, TEM, Confocal: principle and applications.	2
Total		**24**

Course Outcomes:	
After completion of the course, students will be able to:	
1	State different engineering applications from biological perspective.
2	Classify biological systems and identify different organisms and microorganisms depending on their morphological, biochemical and ecological criterion.
3	Explain the concept of recessiveness and dominance during the passage of genetic material from parent to offspring and describe DNA as a genetic material in the molecular basis of information transfer.
4	Discuss structures of different biomolecules starting from basic units and hence understand different biological processes at the reductionistic level.
5	Describe protein structures and enzymology and also compare different mechanisms of enzyme action.
6	Describe energy transformation processes in biological systems.

Learning Resources:	
1	Biology for Engineers. Arthur T. Johnson. CRC Press.
2	Biology and Engineering of Stem Cell Niches. A K Vishwakarma and Jefferey Karp, Elsevier.

3	Environmental Biology for Engineers and Scientists. David A. Vaccari, P. P. Storm and J. F Alleman. ELBS
4	Biology for Engineers. G. K. Suraish Kumar. Oxford

Course Name:	**Engineering Mechanics**		
Course Code:	ES-AUE301	**Category:**	Engineering Science Courses
Semester:	Third	**Credit:**	4
L-T-P:	3-1-0	**Pre-Requisites:**	No-prerequisite
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	The objective of this Course is to provide an introductory treatment of Engineering Mechanics to all the students of engineering, with a view to prepare a good foundation for taking up advanced courses in the area in the subsequent semesters.
2	A working knowledge of statics with emphasis on force equilibrium and free body diagrams provides an understanding of the kinds of stress and deformation and how to determine them in a wide range of simple, practical structural problems, and an understanding of the mechanical behavior of materials under various load conditions.

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Idealization of mechanics, Concept of rigid body, External forces (body forces & surface forces), different support & loading conditions, laws of mechanics. Introduction to vector, vectors and units, vector notation, vector product, Varignon’s theorem and uses.	**6**
2	Force Systems: Basic concepts, Particle equilibrium in 2-D & 3-D; Rigid Body equilibrium; System of Forces, Coplanar Concurrent Forces, Components in Space – Resultant- Moment of Forces and its Application; Couples and Resultant of Force System, Equilibrium of System of Forces, Free body diagrams, Equations of Equilibrium of Coplanar Systems and Spatial Systems; Static Indeterminacy. Friction: Types of friction, Limiting friction, Laws of Friction, Static	**6**

	and Dynamic Friction; Motion of Bodies, wedge friction, screw jack & differential screw jack.	
3	Basic Structural Analysis covering, Equilibrium in three dimensions; Method of Sections; Method of Joints; How to determine if a member is in tension or compression; Simple Trusses; Zero force members; Beams & types of beams; Frames & Machines;	**6**
4	Centroid and Centre of Gravity covering, Centroid of simple figures from first principle, centroid of composite sections, Centre of Gravity and its implications, Area moment of inertia- Definition, Moment of inertia of plane sections from first principles, Theorems of moment of inertia, Moment of inertia of standard sections and composite sections, Mass moment inertia of circular plate, Cylinder, Cone, Sphere, Hook.	**6**
5	Review of particle dynamics- Rectilinear motion; Plane curvilinear motion (rectangular, path, and polar coordinates). 3-D curvilinear motion; Relative and constrained motion; Newton's 2nd law (rectangular, path, and polar coordinates). Work-kinetic energy, power, potential energy. Impulse-momentum (linear, angular); Impact (Direct and oblique). Introduction to Kinetics of Rigid Bodies covering, Basic terms, general principles in dynamics; Types of motion, Instantaneous centre of rotation in plane motion and simple problems; D'Alembert's principle and its applications in plane motion and connected bodies; Work energy principle and its application in plane.	**8**
6	Engineering mechanics of composite materials: basic concept, classification, geometric and physical definition, response under load, elastic and strength behavior.	**4**
Tutorials from the above modules covering, To find the various forces and angles including resultants in various parts of wall crane, roof truss, pipes, etc.; To verify the line of polygon on various forces; To find coefficient of friction between		**12**

various materials on inclined plan; Free body diagrams various systems including block-pulley; To verify the principle of moment in the disc apparatus; Helical block; To draw a load efficiency curve for a screw jack	
Total : 36 Lectures +12 Tutorials	**36L+12T**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Use scalar and vector analytical techniques for analyzing forces in statically determinate structures.
2	Apply fundamental concepts of kinematics and kinetics of particles to the analysis of simple, practical problems.
3	Apply basic knowledge of maths and physics to solve real-world problems
4	Understand basic kinematics concepts – displacement, velocity and acceleration (and their angular counterparts).
5	Understand basic dynamics concepts – force, momentum, work and energy.

Learning Resources:	
1	B. Bhattacharyya, 2014 “Engineering Mechanics”, 2nd Edition, Oxford University Press.
2	F. P. Beer and E. R. Johnston, 2011, “Vector Mechanics for Engineers”, Vol I - Statics, Vol II, – Dynamics, 9th Ed, Tata McGraw Hill.
3	Irving H. Shames, 2006 “Engineering Mechanics”, 4th Edition, Prentice Hall.
4	R. K. Bansal, 2015, “A Text Book of Engineering Mechanics”, 8th Edition, Laxmi Publications.
5	Shanes and Rao, 2006, “Engineering Mechanics”, 4th Edition, Pearson Education.
6	Meriam J.L. and Kraige L.G., 1993 “Engineering Mechanics- Statics – Volume 1, Dynamics”, 3rd Edition, John Wiley & Sons.
7	Timoshenko S.P and Young D.H., 1956 “Engineering Mechanics”, 4th Edition, McGraw Hill.
8	Beer & Johnston, 1987, “Mechanics for Engineers”, 4th Edition, McGraw – Hill.
9	Isaac M. Daniel and Ori Ishai, 2005 “Engineering mechanics of composite materials”, 2nd Edition, Oxford University Press.
10.	Robert M. Jones, 1998 “Mechanics of composite material”, 2nd Edition, CRC Press.

Course Name:	**Fluid Mechanics and Hydraulic Machines**		
Course Code:	ES-AUE302	**Category:**	Engineering Science Courses
Semester:	Third	**Credit:**	4
L-T-P:	4-0-0	**Pre-Requisites:**	No-prerequisite
Full Marks:	100		
Examination	Semester Examination:	Continuous Assessment:	Attendance:

Scheme:	70	25	05

Course Objectives:	
1	To learn about the application of mass and momentum conservation laws for fluid flows.
2	To obtain the velocity and pressure variations in various types of simple flows
3	To understand the importance of dimensional analysis
4	To analyze the flow in water pumps and turbines

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Definition of fluid, Newton's law of viscosity, Properties of fluids, mass density, specific volume, specific gravity, viscosity, compressibility and surface tension. Newtonian vs Non-Newtonian Fluids. Rheological diagram for classification of fluids. Fluid statics; Pascal law and Hydrostatic Law, pressure and pressure measurement. Hydraulic forces on submerged surfaces; forces on vertical, horizontal, inclined and curved surfaces.	**10**
2	Fluid flow and classifications. Continuity equation in 1D & 3D. Potential flow & Stream function; types of flow lines. Motion of a fluid particle, Fluid deformation. Equations of motion; Euler's equation, Bernoulli's equation, Applications of Bernoulli's equation. Momentum Analysis of flow systems; the linear momentum equation for steady flow, differential approach.	**10**
3	Concept of boundary layer, measures of boundary layer thickness, Boundary layer separation. Hagen-Poiseuille equations for friction and pressure drop. Flow of fluid around submerged bodies; basic concepts of drag and lift.	**4**
4	Flow through pipes, Major and minor losses through pipes, Darcy Weisbach equation, friction factor, Moody's diagram. hydraulic gradient line and total energy line. Flow through syphon. Basic principle for flow through orifices, notches & weir (rectangular-v). Flow through open channels; use of Chezy's formula.	**7**
5	Dimensional Analysis & Model investigation applied to flow systems, methods of dimension analysis. Types of similitude. Model analysis. Dimensionless numbers in fluid flow. Similarity laws.	**4**
6	Impact of Jet, Classification of water turbines, heads and efficiencies, velocity triangles, Axial, radial and mixed flow turbines, Pelton wheel, Francis turbine and Kaplan turbines, working principles. Draft tube. Specific speed, unit quantities, performance curves for turbines. Governing of turbines.	**8**

7	Centrifugal pumps: Components, working principle, head & efficiency. Multistage Centrifugal pumps. Pump characteristics, NPSH & Cavitation. Reciprocating Pumps: Components & Principles, Classification, discharge, work done, power requirement.	**5**
Total		**48**

Course Outcomes:	
After completion of the course, students will be able to:	
1	learn about the properties of fluids and analyze the forces in static fluid and forces acting on submerged surfaces.
2	learn about the application of mass and momentum conservation laws for fluid flows.
3	analyze the internal flow in pipes and channels & ability to predict loss of head.
4	understand the importance of dimensional & model analysis
5	analyze the flow in turbines and water pumps

Learning Resources:	
1	A Textbook of Fluid Mechanics and Hydraulic Machines in SI Unit, R.K. Bansal, Laxmi Publication (P) Ltd., 10th Edition, 2018
2	A Textbook of Fluid Mechanics and Hydraulic Machines in SI Unit, R K Rajput, S. Chand & Company Ltd.,6th Edition, 2018
3	Fluid Mechanics and Fluid Power Engineering (SI Units), D S Kumar, S.K. Kataria & Sons, 2013
4	A Textbook on Fluid Mechanics and Machines, S.Pati, McGrawHill, 1st Edition, 2017
5	Introduction to Fluid Mechanics and Fluid Machines, S K Som, Gautam Biswas, and S Chakraborty, McGrawHill, 3rd Edition, 2017
6	Fluid Mechanics and Machinery, C.S.P.Ojha, R. Berndtsson and P. N. Chadramouli, Oxford University Press, 2010
7	Hydraulics and Fluid Mechanics, P M Modi and S M Seth, Standard Book House, 22nd Edition, 2019
8	Fluid Mechanics - Fundamentals and Application, Yunus A Cengel and John M Cimballa, McGrawHill, 4th Edition, 2019

Course Name:	**Applied Thermodynamics**		
Course Code:	PC-AUE301	**Category:**	PC
Semester:	Third	**Credit:**	4
L-T-P:	3-1-0	**Pre-Requisites:**	High school Physics, Mathematics-I(BSM101), Chemistry (BSCH201)
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05
Course Objectives:			

1	To learn about work and heat interactions, and balance of energy between system and its surroundings.
2	To learn about application of I and II laws of thermodynamics for various energy conversion devices.
3	To learn about gas and vapor cycles and their first law and second law efficiencies.
4	To evaluate the changes in properties of substances in various processes.
5	To understand about the refrigeration cycle and air-conditioning system.

Course Contents:

Module No.	Description of Topic	Contact Hrs.
1	**Basic Thermodynamics-I:** Systems − Zeroth low − First law − Steady flow energy equation − Steady Flow Engineering Devices − Energy Balance for Unsteady Flow − Heat and work transfer in flow and non-flow processes − Kelvin's and Clausius statements of second law − Carnot cycle − Carnot Theorem − thermodynamic temperature scale − Carnot heat engine, efficiency − Carnot refrigerator and heat pump.	**7L+3T**
2	**Basic Thermodynamics-II:** Clausius inequality− Definition of entropy − Evaluation of S for solids, liquids, ideal gases and ideal gas mixtures undergoing various processes − Principle of increase of entropy− Illustration of processes in T-S coordinates − Irreversibility and Availability − Availability function for systems and Control volumes undergoing different processes − Lost work − Second law analysis for a control volume − Exergy balance equation and Exergy analysis − Ideal and Real Gases and Thermodynamic Relations− Gas mixtures – properties ideal and real gases − Equation of state− Avogadro's Law − Vander Waal's equation of state − Dalton's law of partial pressure −Exact differentials − T-D relations − Maxwell's relations − Clausius Clapeyron equations − Joule-Thomson coefficient	**10L+2T**
3	**Air Standard Cycle and Compressors:** Otto − Diesel − Dual combustion −Brayton − Atkinson Cycle −Air standard efficiency − Mean effective pressure − Reciprocating compressors.	**5L+2T**
4	**Steam and Jet Propulsion:** Properties of steam − Phase rule, P-V, P-T, T-V, T-S, H-S diagrams, PVT surfaces− Use of steam tables and R134a tables; Saturation tables; Superheated tables; Identification of states & determination of properties − Mollier's chart− Rankine cycle − the ideal reheat and regenerative and the second law analysis of vapour power cycles − Steam Nozzles − Simple jet propulsion system − Thrust rocket motor − Specific impulse.	**8L+3T**
5	**Refrigeration and Air-Conditioning:** Principles of psychometry and refrigeration − Vapour compression − Vapour absorption types − Coefficient of performance − Properties of refrigerants − Basic Principle and types of Air conditioning.	**6L+2T**

Total	**36L+12T**

Course Outcomes:	
After completion of the course, students will be able to:	
1	apply energy balance to systems and control volumes, in situations involving heat and work interactions
2	evaluate changes in thermodynamic properties of pure substances
3	evaluate the performance of energy conversion devices
4	differentiate between high grade and low grade energies

Learning Resources:	
1	Nag. P.K., "Engineering Thermodynamics", 2018, Tata McGraw-Hill, New Delhi.
2	Sonntag, R. E., Borgnakke, C., &Wylen, G. J. V. "Fundamentals of thermodynamic": 2007, John Wiley and Sons.
3	Moran, M. J. and Shapiro, H. N., 1999, Fundamentals of Engineering Thermodynamics, John Wiley and Sons.
4	Jones, J. B. and Duggan, R. E., 1996, Engineering Thermodynamics, Prentice-Hall of India
5	Potter, M. C., & Somerton, C. W. Schaum's Outline of Thermodynamics for Engineers, 2014, McGraw-Hill.

Course Name:	**Manufacturing Methods**		
Course Code:	PC-AUE302	**Category:**	PC
Semester:	Third	**Credit:**	4
L-T-P:	4-0-0	**Pre-Requisites:**	NIL
Full Marks:	100		
Examination Scheme:	Semester Examination: 70	Continuous Assessment: 25	Attendance: 05

Course Objectives:	
1	To impart knowledge and train students with the basic principles, design aspects, defects, applications in area of metal casting, welding and forming processes.

Course Contents:		
Module No.	**Description of Topic**	**Contact Hrs.**
1	Casting: Introduction, Patterns and Pattern making, Moulding sand and its properties, casting design considerations, Melting, pouring and solidification, estimation of pouring and solidification time, Casting processes - Sand, centrifugal, die, investment, lost foam, gravity, squeeze, shell, casting defects and residual stresses.	**12**

2	Joining/fastening processes: Physics of welding, Types of welding, Oxyfuel welding techniques, Arc welding processes- Principles, Characteristics of arc, Manual metal arc welding, Submerged arc welding, Metal inert gas welding, Tungsten inert gas welding, , Plasma arc welding, Resistance welding processes, Thermit welding, Electron beam welding, Laser beam welding; Solid state welding processes- Friction welding, Friction stir welding, Ultrasonic welding; Heat input in welding processes, Defects in welding; Soldering and brazing.	**12**
3	Forming: Introduction to bulk and sheet metal forming, plastic deformation and yield criteria; fundamentals of hot and cold working processes, Basic principles, types and load estimation for bulk forming (forging, rolling, extrusion, drawing) and sheet forming (shearing, deep drawing, bending), Moulding of Thermoplastics – Working principles and typical applications of Injection moulding	**14**
4	Module IV: Powder Metallurgy: Production of metal powders, mixing and blending, compacting, sintering and secondary operations. Application of Powder Metallurgy in Automobile fields.	**4**
5	Overview of Additive manufacturing (AM): Reverse engineering, Different AM processes and relevant process physics, Principles, systems, relative advantages and applications of the common AM methods, such as Stereo lithography, Selective laser sintering, Fused deposition modelling, Laminated objects manufacturing.	**6**
Total		**48**

Course Outcomes:	
After completion of the course, students will be able to:	
1	describe basic principles of casting, welding and forming processes.
2	explain different process parameters and methods of casting, welding and forming processes.
3	solve complex problems of manufacturing using standard methods.
4	design various defect free subsystems of primary manufacturing processes.

Learning Resources:	
1	Mikell P. Groover, Fundamentals of Modern Manufacturing: Materials, Processes, and Systems, Wiley India Publication, 2018.
2	Kalpakjian and Schmid, Manufacturing Processes for Engineering Materials (5th Edition)- Pearson India, 2014.
3	P. N. Rao, Manufacturing Technology - Foundry, Forming and Welding, Volume1 (5th Edition), Tata McGraw Hill Pub,2018.

4	J.T. Black & Ronald. A. Kohser, DeGarmo's Materials and Processes in Manufacturing, SI Version, Wiley India Publication, 2017
5	Sabrie Soloman “3D Printing and Design”, Khanna Publishing House, Delhi,2020
6	A. Ghosh, Rapid Prototyping, EW Press, 2006
7	Chua C.K., Leong K.F. and LIM C.S Rapid prototyping: Principles an Applications, World Scientific publications, 3rdEd., 2010

Course Name:	**Machine Drawing**		
Course Code:	PC –AUE 391	**Category:**	Professional Core courses
Semester:	Third	**Credit:**	1.5
L-T-P:	0-0-3	**Pre-Requisites:**	No prerequisite
Full Marks:	100		
Examination Scheme:	Semester Examination: 60	Continuous Assessment: 35	Attendance: 05

Course Objectives:	
1	Assembly and detailed drawings of a mechanical assembly, such as a simple gear box, flange coupling, welded bracket joined by stud bolt on to a structure, etc.
2	Practicing AutoCAD or similar graphics softwares and making orthographic and isometric projections of different components.

Course Contents:

Module No.	**Description of Topic**	**Contact Hrs.**
1	Schematic product symbols for standard components in mechanical, electrical and electronic systems, welding symbols and pipe joints;	**6**
2	Orthographic projections of machine elements, different sectional views-full, auxiliary sections; Isometric projection of components;	**6**
3	Assembly and detailed drawings of a mechanical assembly, such as a plummer block, tool head of a shaping machine, tail stock of a lathe, simple gearbox,	**6**
4	Flange coupling, welded bracket joined by stud bolt on to a structure, welded pipe joints indicating work parts before welding, etc.	**6**
5	Practicing AutoCAD or similar graphics software and making orthographic and isometric projections of different components.	**12**
Total		**36**

Course Outcomes:	
After completion of the course, students will be able to:	
1	Describe schematic product symbols for standard components in mechanical, electrical and electronic systems, welding symbols and pipe joints.
2	Explain orthographic projections of machine elements, different sectional views and Isometric projection of components.
3	Assemble different machine elements such as a plummer block, tool head of a shaping machine, tailstock of a lathe, welded pipe joints indicating work parts before welding.

Learning Resources:	
1	Text Book on Engineering Drawing, Narayana and Kannaia H, Scitech.
2	Mechanical Engineering Drawing and Design, S. Pal and M. Bhattacharyya.
3	Machine Drawing by N.D. Bhatt.
4	Machine Drawing by P.S. Gill.
5	Engineering Drawing and Graphics + AutoCAD by K. Venugopal, New Age International Pub.
6	Engineering Drawing with an Introduction to AutoCAD by D.A. Jolhe, Tata-McGraw-Hill Co.